

ICMA Bond Data Taxonomy User Guide

XML Schema Definitions

Version 1.2

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Release Table

Release Version	Release Date	Description
V1.0	10 March 2023	Initial release of the BDT data pack.
V1.1	13 July 2023	Typo in enumeration file.
V1.2	24 January 2024 (preliminary)	- DLT extension: - Platform operational information (Key fields and enumerations.) - DLT XML examples. - New party role enumerations. - General, minor enhancements - New fields (eg tranche and series amount, calculation amount, change of interest/payment, issue yield, use of proceeds, redemption payment basis.) - New enumerations (eg for form of note, identifier type, interest type, clearing and settlement system, party roles)

Description of ICMA Bond Data Taxonomy Pack

The ICMA Bond Data Taxonomy XML document defines all the fields covered. The XML definition consists of three XSD files that define the overall structure of a new issuance, enumerations, and ISO data types.

The ICMA BDT Pack consists of the below documents:

- 1) **ICMA Bond Data Taxonomy:** XSD document defining the overall structure of a new issuance and key fields.
- 2) **ICMA Bond Data Taxonomy Enumerations:** XSD document defining the key field enumerations.
- 3) **ICMA Bond Data Taxonomy ISO Types:** XSD document defining the key field ISO Data Types
- 4) **Example files:** Two examples of key bond details from term sheet and final terms documents, represented in XML format as defined by the BDT XSD.
- 5) **ICMA Bond Data Taxonomy - Reference sheet:** Spreadsheet listing key bond data fields together with XML definitions and ISO definitions (where applicable).
- 6) **User Guide**

Background and objectives

The [Bond Data Taxonomy \(BDT\)](#), formerly known as the Common Data Dictionary, was published in March 2023 and provides an agreed language to promote automation and reduce the risk of fragmentation across bond issuance, trading, settlement and distribution.

The BDT is technology agnostic and is designed to be used both for traditional debt securities as well as DLT-based debt securities.

The BDT standardises key economic terms of a vanilla bond (e.g. nominal amounts, denominations, currencies, prices, net proceeds, interest, and interest payment related information), key dates (e.g. pricing, settlement, issue dates) among other information (e.g. whether bearer or registered, relevant parties, ratings, selling restrictions) typically included within a term sheet as initial focus area.

The BDT as a ‘common language’ seeks to:

- Promote straight-through-processing (STP) and interoperability, from issuance, trading, settlement to distribution.
- Be vendor agnostic, facilitating the exchange of data between multiple solutions and systems.
- Lay a common foundation for leveraging new technologies, such as distributed ledger, and developing new services.

Further information is available on the [ICMA BDT webpage](#).

Bond Data Taxonomy video tutorial

A video tutorial on the BDT is available on [ICMA's website](#).

Bond Data Taxonomy in machine-readable format

The BDT provides a consistent terminology and interpretation of expected data attributes, building on existing standards and initiatives, for market participants to map to or reference when exchanging data electronically through the issuance process of a bond. Importantly, the output is represented in a machine-readable format.

ICMA's Bond Data Taxonomy Working Group has developed an agreed composite list of over 100 fields to capture key bond information within the context of a term sheet (see BDT reference sheet), including DLT-related information in version 1.2. Each field is defined by its data type, multiplicity and restrictions that include syntax patterns and predefined enumerations. Many of the fields are defined by ISO standards

such as date, currency and legal identifiers. Based on the data definitions, an Extensible Markup Language (XML) schema was developed and is described in an XML Schema Definition (XSD).

Use of an XML format to represent the ICMA BDT key fields does not preclude the use of other formats.

XML conventions and best practice

- All type names were defined using Upper Camel Case. ISO standards include LEIIdentifier, ISODate, ISOCurrency, MaxText.
- All names are singular except where the plural is important to the domain terminology. Lists of elements would appear with a List or Group prefix or suffix.
- All files use a single name space.
- We focus on tag readability as opposed to acronyms and abbreviations.
- To support human readability all the data types and elements include annotations that define the name and description of the unit.
- Avoided using container names in the element and attribute names except where it was important to the context and domain terminology.
- Enumeration values use full names in upper case with underscore word separators to avoid confusing abbreviations.
- Data type names that are part of a group avoided using contextual names such as IssuerLEI.

Creating and Reading XML

To create an xml document using the bdt xsd, include the following lines in the header of the xml document:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<Document xmlns="urn:icma:xsd:ICMABondDataTaxonomy"  
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
xsi:schemaLocation="urn:icma:xsd:ICMABondDataTaxonomy icma-bond-data-taxonomy.xsd">
```

This assumes the xsd files are in the same directory. In cases where the xsd exist either in a separate directory or on a remote server you should change the file name or url.

Overall Structure

The ICMA Bond Data Taxonomy xml design is defined around four primary data types: PartyRole, Party, Issuance and Product

PartyRole

PartyRoles are defined by the enumerations of PartyRoleType and are used to associate specific parties to a role in the issuance of a new security such as the Issuer, Guarantor, Trustee and Joint Lead Managers. The party roles are restricted through a pre-defined list of enumerations that are contained in the xsd, icma-bond-data-taxonomy-enums.xsd.

The party details of a particular party can be referenced through the reference ID tag, PID in the partyRole element.

```
<PartyRole>
  <PartyRoleType>Issuer</PartyRoleType>
  <PartyID PID="LEI-DG3RU1DBUFHT4ZF9WN62" />
</PartyRole>
```

Party

The Party type holds the details of the party but not its role which allows for a party to perform multiple roles without having to redefine the details of a party. An attribute holds the party identifier that can be referenced from a party role using the reference ID tag, PID. The Party type includes two required elements, PartyName and LEI, and two optional elements, BloombergTicker and PartyRating.

Newly issued securities require ratings of the issuer and rating on the new issue, so the Rating Type is a separate complex type that can hold multiple with elements for RatingAgency, RatingValue, RatingOutlook.

The PartyType can be easily expanded to support more details of the Party such as entity type, industry, and location.

Issuance

The Issuance data type contains all the data elements specifically related to the issuance of a bond including issuance method, issue price, target market, restrictions, place of listing, documentation and governing law. Most of the issuance elements are restricted by pre-defined enumerations. The bond details are defined in the product data type.

There are two issuance types available:

```
<xs:simpleType name="IssuanceType">
```

```

    <xs:restriction base="xs:string">
        <xs:enumeration value="PROGRAMME"/>
        <xs:enumeration value="STANDALONE"/>
    </xs:restriction>
</xs:simpleType>

```

The issue price is defined by a reusable data type called Price that is restricted to a decimal value. A separate data type, PriceYieldDiscount is used for All-in price and All-in yield and the price of the underlying security which may also serve as the benchmark security.

Issuance of new bonds requires definition of selling restrictions, PRIIPs restrictions, clearing systems, method of distribution and governing law. These data elements are all defined under individual data elements and restricted by enumerations: <xs:element name="SellingRestrictions">

```
<xs:element name="PRIIPsRestriction">
```

```
<xs:element name="ClearingSettlementSystem">
```

```
<xs:element name="MethodOfDistribution">
```

```
<xs:element name="GoverningLaw">
```

Additional issuance documentation references can be added in the Documentation element in a xml that allows issuers to include descriptive information about the offering as well as links to offering documents. The number of document references is unbounded.

```

<xs:complexType name="Documentation">
    <xs:complexType name="DocumentDetail">
        <xs:element name="DocumentName">
        <xs:element name="DocumentDate">
        <xs:element name="DocumentLink"

```

If a benchmark security is used it will be defined by the Benchmark tag that is defined as a Security type and will include the identifier and price/yield of the security. The same Security type is reused for the new issue and its benchmark in the case of a floating rate security.

Product

The Product type is intended to be a standalone, reusable definition of a bond and as such includes all the details of the security including identifiers, issued amounts, rating, maturity date, interest payments and calculation conventions.

Product type also defines characteristics of the security that define the seniority (StatusOfNote), registration (FormOfNote), interest rate payout, optionality, and classification (Taxonomy).

The InterestPayment element describes the details of all interest payments of the bond. The payments are defined by the parameters of InterestType, InterestRate, and PaymentDatesAndAmounts that further define the payment frequency and payable dates. An example of this section would be:

```
<InterestPayments>
  <InterestPaymentFrequency>ANNUALLY</InterestPaymentFrequency>
    <PayableDate>
      <Day>30</Day>
      <Month>11</Month>
    </PayableDate>
    <FirstPayment>
      <PaymentDate>2022-11-30</PaymentDate>
      <BrokenDateType>SHORT</BrokenDateType>
    </FirstPayment>
</InterestPayments>
```


DLT Extension

To support the development of DLT-based bond markets, the BDT has been expanded to capture selected DLT-related information of a debt security issuance. The new fields include:

DLT Field	Description	Required/ Optional	Examples
Type	Specifies the name of the underlying platform used in the design and operation of the DLT and covers a range of services such as accessibility, consensus mechanism, and governance. e.g. Ethereum, Hyperledger Besu, Hyperledger Fabric. This is a free text field that users can populate with their specific description.	R	Ethereum, Hyperledger Besu, Hyperledger Fabric
Accessibility	Accessibility describes the scope of participants and whether participants' access is restricted. A public ledger, also known as non-permissioned ledger, allows anyone to create and validate blocks as well as to modify the ledger state by storing and updating data by means of transactions among participating entities. Private ledgers are restricted to specific users or entities to modify the ledger and view transactions on the ledger. Permissioned ledgers are a hybrid of public and private, such that only authorised and trusted entities can view and participate in the activities within the ledger.	O	Public, Private, Permissioned, Consortium, Hybrid, SemiPublic
Role	The Role element describes the activities and services performed by the platform as described in the offering documents. The role type allows for a free text description of one or more roles but also allows the use of a link to a specific role as defined in the partyRole group. Typical roles performed by a DLT platform include registrar, account keeper, custodian and settlement agent.	O	The Notes being registered in a DLT constituting a dispositif d'enregistrement électronique partagé, as defined in Articles L. 211-3 et seq. of the French Monetary and Financial Code (the DEEP).

DLT Field	Description	Required/ Optional	Examples
Fee	A description of the fees (if applicable) charged by the platform, how they are calculated and distributed.	O	One Time 10,000 GBP
Operator	The operator of a DLT platform will depend on the type of platform and accessibility. In a private ledger there may be one operator. In a consortium there will be more than one operator. In a public network, no operator will be defined instead the operator element will be set to Distributed.	R	SG Forge, GSBE, IBERCLEAR
Name	The public or commercial name assigned to the platform.	O	DEEP, HSBC Orion
Identifier	One or more identifiers assigned to the platform that may include a MIC, LEI, DTI or custom identifier.	O	MIC=SGF, DTI=XC7KDF8R9
Token Type	The type of token(s) used on the platform. This is a free text field that allows for multiple tokens.	O	Security,Cash
Token Technical Reference	One or more technical reference documents and specifications of the token(s).	O	ERC-20, ERC-2020
Token Technical Reference ID	Identifiers of the technical specifications reference in the TokenTechnicalReference tag.	O	BD7KDF8R3, XS2524143554
Smart Contract	The smart contract identifier used by the platform in reference to the token type and purpose of the platform.	O	0xE9Ac9Eb7cDDAfd9 e5CA43c23bd521aFc DK8BE49

In the xsd file, the DLT information allows for unlimited number of platforms and is defined in the block:

```
<xs:element name="DltPlatform" type="DltPlatformOperationalInformation" minOccurs="0"
maxOccurs="unbounded"/>
```

DLT xml example:

```
<DltPlatform>
  <Type>Ethereum</Type>
  <Accessibility>Public</Accessibility>
  <Role>Registration</Role>
  <Operator>
    <Operator>LEI-969500FX8K40ZDW4F377</Operator>
  </Operator>
  <Name>DEEP</Name>
  <Identifier PID="LEI-969500FX8K40ZDW4F378">
    <MIC>SGFG</MIC>
  </Identifier>
  <TokenType>SECURITY</TokenType>
  <TokenTechnicalReference>ERC-20</TokenTechnicalReference>
  <SmartContract>0x9915ced2820082a25a8992752aa43bcb0e1d1097</SmartContract>
</DltPlatform>
```

Examples

Also included in the BDT pack are examples of mapping existing issuance documents to a BDT xml document based on the xsd.

Contact us

For any further questions, please contact FinTech@icmagroup.org .